

-- =====

-- **SECTION A : Basic Queries**

-- Database : Celebal_Week2

-- Table : dbo.Superstore

-- Tool : SQL Server Management Studio (SSMS)

-- Author : ARATI THORAT | Celebal Internship – Week 2

-- =====

USE Celebal_Week2;

GO

-- =====

-- **A1. View ALL columns from the Superstore table**

-- PURPOSE : Get a full picture of the raw dataset

-- =====

SELECT *

FROM dbo.Superstore;

OBSERVATION:

The table contains 9,994 rows and 21 columns covering order details, customer info, product info, and financials.

All data is in a single flat/denormalised table.

-- **A2. View only important columns (selective projection)**

-- PURPOSE : Reduce noise, focus on key fields

SELECT

Row_ID,
Order_ID,
Order_Date,
Ship_Date,
Ship_Mode,
Customer_Name,
Segment,
Region,
Category,
Sub_Category,
Product_Name,
Sales,
Quantity,
Discount,
Profit

FROM dbo.Superstore;

OBSERVATION:

Projecting only the business-relevant columns makes the result easier to read and speeds up query execution.

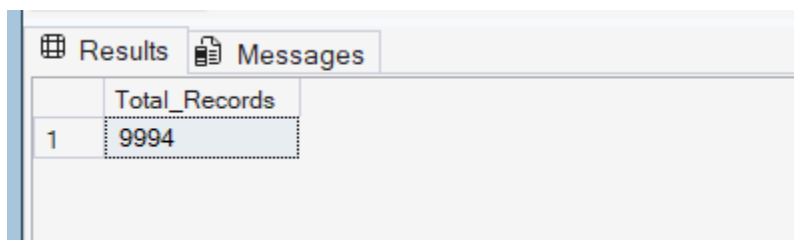
-- **A3. Total row count**

-- PURPOSE : Confirm data load was complete

```
SELECT COUNT(*) AS Total_Records
```

```
FROM dbo.Superstore;
```

Result :



The screenshot shows a SQL Server query results window. It has two tabs: 'Results' (active) and 'Messages'. The 'Results' tab displays a single row of data. The first column is labeled 'Total_Records' and contains the value '9994'. The row is highlighted with a blue background.

Total_Records
9994

OBSERVATION:

Expected result → 9,994 rows.

If any number other than 9,994 appears, the CSV import may have been incomplete and needs to be re-imported

-- **A4. Inspect column names and data types from schema**

-- PURPOSE : Understand the table structure before querying

```
SELECT
```

```
    COLUMN_NAME,
```

```
    DATA_TYPE,
```

```
    CHARACTER_MAXIMUM_LENGTH,
```

```
    IS_NULLABLE
```

```
FROM INFORMATION_SCHEMA.COLUMNS
```

WHERE TABLE_NAME = 'Superstore'

AND TABLE_SCHEMA = 'dbo'

ORDER BY ORDINAL_POSITION;

Result :

	COLUMN_NAME	DATA_TYPE	CHARACTER_MAXIMUM_LENGTH	IS_NULLABLE
1	Row_ID	smallint	NULL	NO
2	Order_ID	nvarchar	50	NO
3	Order_Date	date	NULL	NO
4	Ship_Date	date	NULL	NO
5	Ship_Mode	nvarchar	50	NO
6	Customer_ID	nvarchar	50	NO
7	Customer_Name	nvarchar	50	NO
8	Segment	nvarchar	50	NO
9	Country	nvarchar	50	NO
10	City	nvarchar	50	NO
11	State	nvarchar	50	NO
12	Postal_Code	int	NULL	YES
13	Region	nvarchar	50	NO
14	Product_ID	nvarchar	50	NO
15	Category	nvarchar	50	NO
16	Sub_Category	nvarchar	50	NO
17	Product_Name	nvarchar	150	NO
18	Sales	float	NULL	NO
19	Quantity	tinyint	NULL	NO
20	Discount	float	NULL	YES
21	Profit	float	NULL	YES

OBSERVATION:

Key columns and their expected types:

Order_Date → date / varchar (check if parsed correctly)

Sales → float / decimal

Profit → float / decimal

Quantity → int

Discount → float

If dates are stored as VARCHAR instead of DATE,

conversion functions (CONVERT / TRY_CONVERT) must be used.

-- **A5. DISTINCT values in categorical columns**

-- PURPOSE : Understand the domain of key dimensions

-- Unique customer segments

```
SELECT DISTINCT Segment FROM dbo.Superstore;
```

Results		Messages	
	Segment		
1	Corporate		
2	Home Office		
3	Consumer		

-- Unique regions

```
SELECT DISTINCT Region FROM dbo.Superstore;
```

Results		Messages	
	Region		
1	East		
2	South		
3	West		
4	Central		

-- Unique ship modes

```
SELECT DISTINCT Ship_Mode FROM dbo.Superstore;
```

Results		Messages
	Ship_Mode	
1	First Class	
2	Same Day	
3	Standard Class	
4	Second Class	

-- Unique product categories

```
SELECT DISTINCT Category FROM dbo.Superstore;
```

Results		Messages
	Category	
1	Office Supplies	
2	Furniture	
3	Technology	

-- Unique sub-categories

```
SELECT DISTINCT Sub_Category
```

```
FROM   dbo.Superstore
```

```
ORDER BY Sub_Category;
```

Results		Messages
	Sub_Category	
1	Accessories	
2	Appliances	
3	Art	
4	Binders	
5	Bookcases	
6	Chairs	
7	Copiers	
8	Envelopes	
9	Fasteners	
10	Furnishings	
11	Labels	
12	Machines	
13	Paper	
14	Phones	
15	Storage	
16	Supplies	
17	Tables	

-- Unique countries

```
SELECT DISTINCT Country FROM dbo.Superstore;
```

Results		Messages
	Country	
1	United States	

OBSERVATION:

Segments : Consumer, Corporate, Home Office

Regions : East, West, Central, South

Ship Modes : Standard Class, Second Class, First Class, Same Day

Categories : Furniture, Office Supplies, Technology

Sub-Cats : 17 unique sub-categories

Country : United States only (domestic dataset)

-- A6. First 10 rows (quick data preview)

-- PURPOSE : Sanity-check the data after import--

SELECT TOP 10 *

FROM dbo.Superstore;

Results		Messages																
	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City	State	Postal_Code	Region	Product_ID	Category	Sub_Category	Product_Name	Sales
1	1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gule	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.95999145
2	2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gule	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Ro...	731.94000244
3	3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by Univ...	14.619999885
4	4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.57751464
5	5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	22.368000030
6	6	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Accessor...	48.860000610
7	7	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322	7.2800002098
8	8	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone	907.15197753
9	9	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by Sam...	18.503999710
10	10	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge	114.90000152

City	State	Postal_Code	Region	Product_ID	Category	Sub_Category	Product_Name	Sales	Quantity	Discount	Profit
Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.959991455078	2	0	41.9136009216309
Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Ro...	731.940002441406	3	0	219.582000732422
Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by Univ...	14.6199998855591	2	0	6.87139987945557
Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.577514648438	5	0.449999988079071	-383.031005859375
Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	22.3680000305176	2	0.200000002980232	2.51640009880066
Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Accessor...	48.8600006103516	7	0	14.1694002151489
Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322	7.28000020980835	4	0	1.96560001373291
Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone	907.151977539063	6	0.200000002980232	90.715202331543
Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by Sam...	18.503999710083	3	0.200000002980232	5.78249979019165
Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge	114.900001525879	5	0	34.4700012207031

OBSERVATION:

The first 10 rows give a quick look at data formatting.

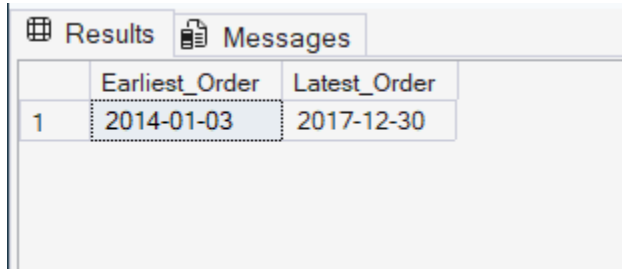
Check that dates, decimals, and text fields look correct.

-- A7. Date range of orders

-- PURPOSE : Understand the time span of the dataset

SELECT


```
MIN(Order_Date) AS Earliest_Order,  
MAX(Order_Date) AS Latest_Order  
FROM dbo.Superstore;
```



	Earliest_Order	Latest_Order
1	2014-01-03	2017-12-30

OBSERVATION:

Dataset spans from 2014 to 2017 (4 years of sales data).

This enables multi-year trend analysis.

-- A8. Sales and Profit range

-- PURPOSE : Understand the financial spread of the data

SELECT

```
ROUND(MIN(Sales), 2) AS Min_Sale,  
ROUND(MAX(Sales), 2) AS Max_Sale,  
ROUND(AVG(Sales), 2) AS Avg_Sale,  
ROUND(MIN(Profit), 2) AS Min_Profit,  
ROUND(MAX(Profit), 2) AS Max_Profit,  
ROUND(AVG(Profit), 2) AS Avg_Profit  
FROM dbo.Superstore;
```

Results		Messages				
	Min_Sale	Max_Sale	Avg_Sale	Min_Profit	Max_Profit	Avg_Profit
1	0.44	22638.48	229.86	-6599.98	8399.98	28.7

OBSERVATION:

- Min Sale ≈ \$0.44 | Max Sale ≈ \$22,638
- Some Profit values are NEGATIVE → loss-making transactions exist.
- The wide range suggests outlier orders worth investigating.

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-- A9. NULL / missing value check

-- PURPOSE : Identify data quality issues early

--

SELECT

```

SUM(CASE WHEN Row_ID IS NULL THEN 1 ELSE 0 END) AS Null_RowID,
SUM(CASE WHEN Order_ID IS NULL THEN 1 ELSE 0 END) AS Null_OrderID,
SUM(CASE WHEN Customer_Name IS NULL THEN 1 ELSE 0 END) AS Null_CustomerName,
SUM(CASE WHEN Sales IS NULL THEN 1 ELSE 0 END) AS Null_Sales,
SUM(CASE WHEN Profit IS NULL THEN 1 ELSE 0 END) AS Null_Profit,
SUM(CASE WHEN Quantity IS NULL THEN 1 ELSE 0 END) AS Null_Quantity,
SUM(CASE WHEN Discount IS NULL THEN 1 ELSE 0 END) AS Null_Discount,
SUM(CASE WHEN Region IS NULL THEN 1 ELSE 0 END) AS Null_Region,
SUM(CASE WHEN Category IS NULL THEN 1 ELSE 0 END) AS Null_Category

```

FROM dbo.Superstore;

Results

Messages

	Null_RowID	Null_OrderID	Null_CustomerName	Null_Sales	Null_Profit	Null_Quantity	Null_Discount	Null_Region	Null_Category
1	0	0	0	0	1	0	0	0	0

OBSERVATION:

All critical columns should return 0 nulls.

If any null count > 0, those records need cleaning

before running aggregation queries.

-- A10. Duplicate Row_ID check

-- PURPOSE : Verify primary key uniqueness after import

SELECT

Row_ID,

COUNT(*) AS Occurrence_Count

FROM dbo.Superstore

GROUP BY Row_ID

HAVING COUNT(*) > 1;

Results		Messages	
	Row_ID	Occurrence_Count	

OBSERVATION:

An empty result = no duplicates = clean import.

If duplicates exist, the CSV import included the header
row twice or had repeated rows — needs cleaning.

-- **A11. Count of unique customers, products, and orders**

-- PURPOSE : Understand dataset cardinality

SELECT

```
COUNT(DISTINCT Customer_ID) AS Unique_Customers,  
COUNT(DISTINCT Product_ID) AS Unique_Products,  
COUNT(DISTINCT Order_ID) AS Unique_Orders,  
COUNT(DISTINCT State) AS Unique_States,  
COUNT(DISTINCT City) AS Unique_Cities
```

FROM dbo.Superstore;

Results		Messages			
	Unique_Customers	Unique_Products	Unique_Orders	Unique_States	Unique_Cities
1	793	1862	5009	49	531

OBSERVATION:

~793 unique customers

~1,862 unique products

~5,009 unique orders

49 states covered

531 cities